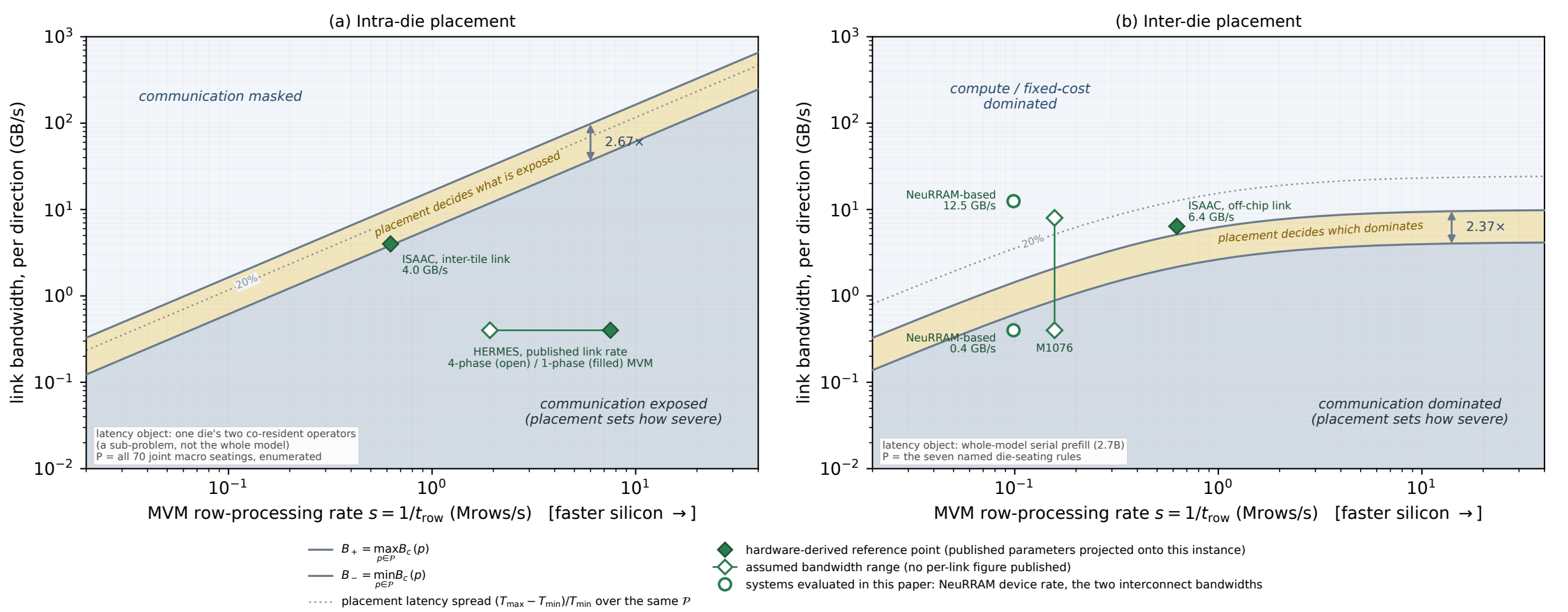
(a) Intra-die placement
communication masked
placement decides what is exposed
2.67×
20%
ISAAC, inter-tile link
4.0 GB/s
HERMES, published link rate
4-phase (open) / 1-phase (filled) MVM
communication exposed
(placement sets how severe)
latency object: one die's two co-resident operators
(a sub-problem, not the whole model)
P = all 70 joint macro seatings, enumerated
link bandwidth, per direction (GB/s)
MVM row-processing rate $s = 1/t_{\text{row}}$ (Mrows/s) [faster silicon →]
(b) Inter-die placement
compute / fixed-cost
dominated
NeuRRAM-based
12.5 GB/s
ISAAC, off-chip link
6.4 GB/s
placement decides which dominates
2.37×
20%
NeuRRAM-based
0.4 GB/s
M1076
communication dominated
(placement sets how severe)
latency object: whole-model serial prefill (2.7B)
P = the seven named die-seating rules
link bandwidth, per direction (GB/s)
MVM row-processing rate $s = 1/t_{\text{row}}$ (Mrows/s) [faster silicon →]
$B_+ = \max_{p \in \mathcal{P}} B_c(p)$
$B_- = \min_{p \in \mathcal{P}} B_c(p)$
placement latency spread $(T_{\max} - T_{\min})/T_{\min}$ over the same $\mathcal{P}$
hardware-derived reference point (published parameters projected onto this instance)
assumed bandwidth range (no per-link figure published)
systems evaluated in this paper: NeuRRAM device rate, the two interconnect bandwidths